Polynomial Centers for Diophantine Consistency

A degree-208 equation in ten positive variables,
with direct quartic companions in 203 naturals and 233 integers

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Abstract

We give a literal integer polynomial in ten strictly positive variables, of exact total degree 208, evaluated by 116 live binary arithmetic operations. Its unsolvability is equivalent to Con(ZFC) by an ordinary proof formalizable in ZFC, composed with explicitly identified inherited arithmetic and refutation calibrations. The new step centers the whole Pell product gV, using polynomial division of EV and a truncated square-root expansion. All divisions disappear from the final formula. A signed majorant argument establishes soundness before any positivity conclusion about the encoded computation is used. This also resolves the previously unproved square-transport case in source 163.

The package includes smaller-operation alternatives at degree 232, 228, 224 and 220, with the same ten positive variables and the same entire fixed coefficient tuple. A separate direct-calculus component gives fully expanded quartics in 203 nonnegative variables and 233 unrestricted integers. The direct component does not simulate K215 or use Jones's universal equation. No global minimum, proof of consistency, or new end-to-end proof-assistant certificate is claimed.

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1 Result and comparison boundary

The objective is an actual polynomial with fixed integer coefficients, not an exponential Diophantine formula or an unspecified instance of MRDP. Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$. The domain is part of every assertion.

Theorem 1.1 (Main equation)**.** *Let P_{208} be the polynomial defined in Sections 2 and 3, with the four fixed integers defined in Appendix A. Then $P_{208} \in \mathbb{Z}[J, F, \alpha, i, T, w, s, n, \eta, \zeta]$, its exact total degree is 208, and*

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P_{208}(\mathbf{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}).$$

The displayed provability claim is established by the ordinary proofs here and the precisely named inherited interfaces. This package does not contain a newly checked object-language ZFC derivation or a newly compiled end-to-end Lean or Rocq development.

We use the immutable repository snapshot

`fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb.`

The comparison includes recent packages 163–165, not merely the older summary table. Source 163 gives ten-positive degree 232 in 87 operations; source 164 gives degree 304 in 85 operations [1, 2]. Their two mechanisms compose to give 85/232. The further centers below give new degree/cost choices.

Equation	Operations	Exact degree	Main change
Source 164	85	304	Inverse Lucas correction
Source 163	87	232	Unsquared strong trace
Trace plus inverse	85	232	Composition of the two
Lower center	87	228	$(A-1)UE$
Center at AUE	88	224	First norm cancellation
Quotient center	96	220	Divide E by A
Whole-product center	**116**	**208**	**Divide EV by powers of A**

All these counts concern ten independent positive variables and the same whole coefficient tuple. Every live binary addition, subtraction and multiplication is charged, including squares and fixed scalar products. Named values are polynomial abbreviations, not extra unknowns. These operation counts are not bit-complexity estimates: the largest fixed coefficient port has 592,884 bits.

Theorem 1.2 (Direct quartic companions)**.** *The literal residual lists in the package define quartics*

$$P_{203}^{\mathbb{N}} = \sum_{j=0}^{154} q_j^2, \qquad P_{233}^{\mathbb{Z}} = \sum_{j=0}^{168} r_j^2,$$

in respectively 203 nonnegative and 233 unrestricted integer variables, such that ZFC proves each no-zero statement equivalent to $\mathrm{Con}(\mathsf{ZFC})$*. All q_j, r_j are explicitly listed integer quadratics; complete collected quartic coefficients are also supplied. These constructions use the numerical proof calculus C222 and its ZF/ZFC calibration.*

This is a set of measured improvements, not a claim that one ordering of "simplicity" dominates every other. The repository already has fewer-variable equations of much larger degree. Coefficient size, monomial count, proof burden, variable domain, degree and evaluation cost must be kept separate.

The ten-positive-variable equation immediately becomes a ten-nonnegative-variable equation of the same degree by replacing each input x_j with $u_j + 1$. If an unrestricted integer domain is required, the four-square theorem gives the completely explicit alternative

$$P_{208}\left(1+\sum_{r=1}^{4} z_{1r}^2, \ldots, 1+\sum_{r=1}^{4} z_{10,r}^2\right) = 0, \qquad \mathbf{z} \in \mathbb{Z}^{40},$$

of exact degree 416. Each positive input has such a representation. The degree is exact because restricting each block to one square leaves the nonzero leading polynomial with every exponent doubled. This standard domain conversion is separate from the direct quartic reductions.

2 The complete common polynomial definitions

Quantify exactly

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed integers are β, K_0, M_C, M_F. They are supplied both as authenticated binary numerals and by the complete finite recipe in Appendix A. In particular,

$$\beta = 2^{590069} - 1, \qquad 0 < M_C < \beta < M_F < 2\beta, \qquad K_0 > 0.$$

These inequalities alone do not calibrate an arbitrary four-tuple. The construction uses all four actual integers together.

Define the following ordinary polynomials in the ten variables:

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha,\\
X &= qw, & Y &= qs, & k &= \eta + \zeta,\\
c &= kY + \eta, & A &= Y(X+1) + 1, & \Delta &= A^2 - 1,\\
H &= 4A - 5, & \ell &= (XY)(kY), & E &= 1 + \ell(\ell + k),\\
R &= [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, & & & &\\
B &= (K_0 + w)C - F, & U &= ic^2, & a &= \Delta U^2 E,\\
F_0 &= 1 + a, & G &= EF_0, & f &= 1 + 2a,\\
Q &= 4\Delta a F_0, & b &= cT - R, & V &= fb - c,\\
K &= X(3c - 2X) + 2, & D &= 6q^6, & M &= hDH,\\
e_6 &= 6(k - R - 1) - \ell(k^2 - 4). & & & &
\end{aligned}$$

The letters C, F, F_0, G, H, K here have only the meanings displayed above. They are not coefficients to be chosen or additional quantifiers.

For any polynomial shift Ω and positive fixed integer r, put

$$Z_0 = hD\{H(n - b^2B) + K\} - H\{DB + h(e_6 - 6q^4 s)\}, \tag{1}$$

$$Z = Z_0 - M\Omega, \tag{2}$$

$$P_{\Omega,r} = Q(Z^2 - r^2M^2GV^2) + (r^2M^2G - Z^2). \tag{3}$$

The principal equation uses $r = 4$ and the explicitly polynomial shift in the next section. Equivalently, its literal evaluator inserts $n - b^2B - \Omega$ inside the braces in (1), saving one operation.

Every exponent in these definitions is a fixed nonnegative integer. In particular, q^6 denotes an ordinary polynomial, evaluated by repeated multiplication. There is no variable exponent, floor, factorial or unbounded product in $P_{\Omega,r}$.

3 A division-free center for the whole product

The following short formula completely specifies the new shift. Put

$$\begin{aligned} e_2 &= (kY)^2, \\ S_E &= (kY)\{\ell + k - (kY)(Y+1)\}, \\ \Theta &= U^2 b, \\ R_E &= 3E + e_2, \\ H_E &= 4E^2 + 3S_E^2 + e_2^2, \end{aligned}$$

$$\boxed{\Omega = U\left[2S_E(\Theta R_E - b + c) + A(4EV - \Theta H_E)\right].} \tag{4}$$

Finally, the requested equation is

$$\boxed{P_{208} = Q(Z^2 - 16M^2GV^2) + (16M^2G - Z^2) = 0.} \tag{5}$$

Together with the finite coefficient recipe, these displayed definitions are one complete polynomial specification in ten positive variables. The supplied straight-line graph is a second, machine-readable specification of exactly this polynomial.

3.1 Where the formula comes from

Introduce a formal indeterminate t, for the proof only, and write

$$\begin{aligned} \mathcal{E}(t) &= e_2 t^2 + e_1 t + e_0, \\ e_1 &= k^2 Y(1 - 2Y - 2Y^2), \\ e_0 &= 1 + k^2 Y(Y+1)\{Y(Y+1) - 1\}, \\ \mathcal{V}(t) &= \{1 + 2(t^2 - 1)U^2 \mathcal{E}(t)\}b - c, \\ \mathcal{F}(t) &= \mathcal{E}(t)\mathcal{V}(t). \end{aligned}$$

Then $\mathcal{E}(A) = E$, $\mathcal{V}(A) = V$ and $\mathcal{F}(A) = EV$. When taking polynomial quotients, Y, k, U, b, c are held as coefficient parameters. Their eventual substitutions as polynomials in the original ten variables do not invalidate a polynomial identity.

For $j = 1, 3$, let Q_j be the value at A of the ordinary polynomial quotient of $\mathcal{F}(t)$ by t^j. Since $\mathcal{E}(t)(b - c)$ has degree two and the other summand of $\mathcal{F}(t)$ is $2U^2 b(t^2 - 1)\mathcal{E}(t)^2$, every coefficient of Q_3 is even. Thus the polynomial

$$U\left(4AEV - 2Q_1 - \frac{Q_3}{2}\right) \tag{6}$$

has integer coefficients. Direct polynomial multiplication proves that it is exactly (4). The slash in (6) is explanatory notation only; equation (4) contains no division operation.

The identity can also be checked without expanding a large expression. Let $S = S_E$, $J_E = S(E + e_0)$ and $D_E = e_2(2S - Ae_2)$. Since $E = AS + e_0$,

$$Q_1 = S(b - c) + 2\Theta(AE^2 - J_E), \qquad \frac{Q_3}{2} = \Theta(J_E - D_E).$$

Substitution into (6) gives

$$\Omega = U\{4AEV - 2S(b - c) - \Theta\mathcal{H}\}, \quad \mathcal{H} = A(4E^2 + 3S^2 + e_2^2) - 2S(3E + e_2),$$

and hence (4).

The formal expansion

$$\sqrt{t^2-1} = t - \frac{1}{2t} - \frac{1}{8t^3} + O(t^{-5})$$

suggests the center, but is not used as an analytic assumption in the Diophantine proof. All required cancellations are exact polynomial identities. The decisive change is to take the polynomial part of the *whole* expression $U\mathcal{F}(t)\sqrt{t^2-1}$, instead of first approximating g and then multiplying that approximation by V.

4 Soundness before the signs of the encoded data are known

This section states the inherited interfaces and proves the new signed center estimate. The detailed parent arguments are retained in `compact/parents/`, including its nested `proof_sources/`; their canonical source paths and hashes are in the source ledger. The original unbounded arguments are not replaced by finite numerical tests.

4.1 Root extraction and the low-index exclusions

At every positive input, independently of the equation,

$$q \ge 32, \quad X, Y \ge q, \quad k \ge 2, \quad c > kY, \quad U \ge c^2, \quad a > c, \quad f > 2c, \quad M > 0.$$

Moreover,

$$f^2 = 1 + 4aF_0, \qquad Q = \Delta(f^2-1) = 4\Delta^2U^2G, \qquad f - 1 = 2\Delta i^2c^4E. \tag{7}$$

If $b = 0$, then $V = -c$; otherwise $|V| \ge f - c > 1$. At a zero of (5),

$$(Q-1)Z^2 = 16M^2G(QV^2-1).$$

Since Q is a multiple of G, both $Q-1$ and QV^2-1 are coprime to G. Prime valuations show that G is a square. The fixed factor 16 and the factor M^2 contribute only even valuations. Also $\gcd(E, F_0) = 1$, so

$$E = \tau^2, \quad F_0 = f_0^2, \quad G = g^2, \qquad g = \tau f_0 > 0.$$

The adjacent-coefficient Pell descent from the parent, applied initially over the rationals, gives an integer $y > 0$ such that

$$QV^2 - (Q-1)y^2 = 1, \qquad Z/M = \varepsilon\, 4gy, \quad \varepsilon \in \{-1, 1\}. \tag{8}$$

For completeness, if the rational ordinate has reduced denominator m, then $m^2 \mid Q-1$. The norm-preserving transformation

$$u' = (2Q-1)u - 2(Q-1)y, \qquad y' = (2Q-1)y - 2Qu$$

strictly decreases $|u| > 1$ after sign reflection, keeps its first coordinate integral, and preserves the reduced denominator of the second. It ends at $u = 1, y = 1$, proving $m = 1$.

Put $I = 2Ug$. The signed all-index folding theorem applies to

$$f^2 - \Delta I^2 = 1, \quad Q = (\Delta I)^2, \quad c^2 \mid I, \quad V \equiv -c \pmod{f}.$$

It yields $c = \psi_A(p)$ for an odd positive p, where $\psi_A(0) = 0$, $\psi_A(1) = 1$ and $\psi_A(j+2) = 2A\psi_A(j+1) - \psi_A(j)$. The first norm yields $k = 2\psi_{1+2XY^2}(m)$ for $m \geq 1$. The parent excludes $m = 1$ and $p = 3$, giving

$$c > 8XY^3 \geq 8q^4, \quad c > X, \quad K > 0, \quad c > 4\Delta + 3.$$

These steps use no sign assumption about B, C, R, b.

The adjacent auxiliary index is odd. Index one would give $|V| = 1$. Index three would give one of the two impossible divisibilities

$$f(b - 4\Delta f) = c - (4\Delta + 3), \qquad f(b + 4\Delta f) = c + (4\Delta + 3),$$

whose right sides lie strictly between zero and f. Thus its index is at least five. Also $b = 0$ is impossible, since then $y^2 - c^2 = (c^2 - 1)/(Q - 1)$ lies strictly between zero and one. We obtain the strong absolute estimates

$$\operatorname{sgn}(V) = \operatorname{sgn}(b), \quad |V| > 8Q^2, \quad |b| > 8f^3 - \tfrac{1}{2} > 7f^3, \tag{9}$$

$$g < f/6, \quad y < \tfrac{3}{2}|V|, \quad |V| < \tfrac{3}{2}f|b|, \quad AUE < \tfrac{3}{2}g < f/4. \tag{10}$$

The bound $g < f/6$ follows from $a^2/g^2 = \Delta U^2 a/(1+a) > 9$; the last bound uses $G = E + \Delta(UE)^2$.

4.2 A uniform polynomial-majorant lemma

Lemma 4.1 (The whole-product center is controlled)**.** *Under the preceding positive-input hypotheses and the consequences of the Pell equations, before recovering the sign of B, the polynomial Ω has the sign of V and satisfies*

$$3AUE|V| < |\Omega| < 5AUE|V|.$$

Proof. For a polynomial in the formal variable t, let a superscript $\#$ denote the sum obtained by taking absolute values of its coefficients and then evaluating at the positive real number A. This is proof notation, not part of the Diophantine expression.

Put $\delta = (Y+1)/A$. Since $X, Y \geq 32$, $\delta < 1/16$. We have

$$|e_1| < 2e_2(Y+1), \qquad 0 < e_0 \leq e_2(Y+1)^2.$$

Consequently

$$\mathcal{E}^{\#}(A) \leq e_2(A+Y+1)^2, \qquad E \geq e_2A^2(1-2\delta),$$

so $\mathcal{E}^{\#}(A) < 2E$.

Using $\mathcal{F}(t) = \mathcal{E}(t)(b-c) + 2U^2b(t^2-1)\mathcal{E}(t)^2$ and $|b| \geq 1$ gives

$$\mathcal{F}^{\#}(A) < 2E(|b| + c) + 8(A^2+1)U^2E^2|b|.$$

Here $1 + c < 2U^2E$ and $|V| > (f-c)|b| > \Delta U^2E|b|$. Therefore

$$\frac{\mathcal{F}^{\#}(A)}{|EV|} < \frac{4}{\Delta} + \frac{8(A^2+1)}{\Delta} = 8 + \frac{20}{\Delta} < 10. \tag{11}$$

Deleting lower powers in a polynomial gives $|Q_j| \leq \mathcal{F}^{\#}(A)/A^j$. Equation (6) therefore implies

$$|\Omega - 4AUEV| < 10U|EV|\left(\frac{2}{A} + \frac{1}{2A^3}\right) < AU|EV|,$$

because $A \geq 16$ and $20/A^2 + 5/A^4 < 1$. This proves both the strict sign assertion and the two bounds. □

Corollary 4.2 (The square transport budget)**.** *Before assuming* $B \geq 0$*,*

$$4gy + |\Omega| < b^2/4.$$

Proof. By (10) and Lemma 4.1,

$$4gy + |\Omega| < 6g|V| + 5AUE|V| < \frac{9}{4}f|V| < \frac{27}{8}f^2|b|.$$

Since $|b| > 7f^3$ and $f > 2$, the ratio of the last bound to b^2 is less than $27/(56f) < 1/4$. □

4.3 Recovering the semantic signs without a circular argument

For proof only, put

$$\rho = \frac{k - R - 1 - XY^2k(k^2-4)/6}{q^6}, \quad \sigma = \frac{s}{q^2}, \quad t_0 = \frac{B}{h}, \quad \kappa = \frac{K}{H}, \quad W = 1 + hb^2.$$

The exact rational carrier identity is

$$Z/M = n + \kappa + \sigma - Wt_0 - \rho - \Omega. \tag{12}$$

The inverse Lucas correction is crucial: $k \geq 2$ implies

$$k - R - 1 - XY^2k(k^2-4)/6 \leq k - R - 1 < b.$$

Thus $\rho < |b|/q^6 < b^2/2$, uniformly for positive, zero or negative R. If $B \leq -1$, then $Wt_0 < -b^2$. Since $\kappa, \sigma > 0$, (8), (12) and Corollary 4.2 would give

$$n \leq 4gy + |\Omega| - \kappa - \sigma + Wt_0 + \rho < \frac{b^2}{4} - b^2 + \frac{b^2}{2} < 0,$$

contradicting $n > 0$. Therefore $B \geq 0$, and its defining equation forces $C > 0$.

Only now may the calibrated mask estimate $3q < R < q^4 - 2$ be used. It gives $c > 2R$ and hence $b, V > 0$. The normalized-rank theorem and the congruences

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}$$

identify $p = R$. The second congruence uses $f \equiv 1 \pmod{c}$ from (7), not an assumed sign. The old signed cases are thereby covered before the positive computation is reconstructed.

5 Index recovery, completeness, and the logical endpoint

5.1 The inverse Lucas correction retains its exact denominator order

Because Ω is an integer polynomial, $Z_0/M = \varepsilon 4gy + \Omega$ is integral. Thus the parent inverse-correction arithmetic applies to the unshifted carrier with exactly the same divisibility conditions. For clarity, the essential order is summarized here.

The recurrence expansion, with $v = XY^2$, is

$$k = 2\psi_{1+2v}(m) = 2m + 8v\binom{m+1}{3} + 32v^2Z_m, \qquad Z_m \in \mathbb{N}.$$

Writing $d = 2m - R - 1$ and $e^* = k - R - 1 - vk(k^2-4)/6$, its exact cubic-difference identity gives

$$3(e^* - d) = 4v^2\{24Z_m - A_m S_m\},$$

where $A_m = \binom{m+1}{3} + 4vZ_m$ and $S_m = k^2 + k(2m) + (2m)^2 - 4$. Since $v = q^3 w s^2$, this proves $q^4 \mid (e^* - d)$ even at the prime 3. Integral Z_0/M also gives $q^4 \mid e^*$. The first Pell base exceeds A, so $m < R$; the bound $R < q^4$ now forces $d = 0$.

One must not yet cancel the factor 3 at the full q^6 level. First reduce the literal numerator modulo H to obtain $H \mid 6hK$. The inherited uniform ratio bound and whole-product projection force $X = 2^R$, without canceling arbitrary factors of a congruence. As $q \mid X$, q is a power of two. Only then does the identity above give $q^6 \mid e^*$. It follows that $q^2 \mid s$, $q^3 \mid Y$, $\kappa \in \mathbb{Z}_{>0}$ and $h \mid B$. These are exactly the native outer conditions of the pinned calibration. This preserves the explicit third-primary repair in source 164.

5.2 Constructing positive witnesses from a refutation

Conversely, a finite syntactic ZFC refutation supplies the parent's canonical outer data with

$$X = 2^R, \quad q^3 \mid Y, \quad C > 0, \quad t_0 = B/h \in \mathbb{N}, \quad \rho, \sigma, \kappa \in \mathbb{Z}.$$

The retained Pell-period construction gives positive i, η, ζ, T and positive V, y satisfying the Pell equation in (8). For the unsquared trace, the old auxiliary coordinate changes by the explicitly integral formula

$$T_{\text{new}} = fT_{\text{old}} - \frac{f-1}{c}R = \frac{V + c + Rf}{cf} > 0.$$

The quotient $(f-1)/c = 2\Delta i^2 c^3 E$ is a polynomial. The new center obeys $\Omega > 0$. Define

$$n = 4gy + \Omega + (1 + hb^2)t_0 + \rho - \kappa - \sigma. \tag{13}$$

Every term is an integer. The parent direct estimates give $g > \kappa + \sigma + |\rho|$, including the potentially negative inverse Lucas correction. Hence $n > 0$, the carrier equals $4gy$, and the literal polynomial vanishes. This constructs all ten positive witnesses; it does not assert that a refutation or a zero actually exists.

5.3 What the equivalence means internally

The fixed tuple is the indexed native compilation of the calibrated machine used by the pinned parents. Their outer reconstruction connects a positive zero to a finite syntactic refutation. Composing the two directions above gives, inside ZFC,

$$\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P_{208}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists p\ \text{Proof}_{\mathsf{ZFC}}(p, \bot).$$

Negating proves Theorem 1.1. This reasoning uses finite coding, integer arithmetic, recurrence identities and the specified inherited proof transformations, all formalizable in ZFC. It is not an inference from agreement only on standard numerals. Nor does it select coefficients according to the unknown truth value of Con(ZFC).

The direct quartics instead use the named interfaces `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The new Diophantine transformations are ordinary mathematical proofs; the inherited Lean/Rocq developments were not rebuilt in this session.

6 Exact degree and the literal arithmetic count

The cancellation giving degree 208 must be checked on the entire polynomial. Propagating degrees through the original squared expression alone would give only an upper bound and miss the main result.

Assign the intermediate weights

$$\deg A = 4, \quad \deg Y = 2, \quad \deg k = 1, \quad \deg U = 7, \quad \deg b = 4, \quad \deg c = 3.$$

An independent sparse expansion of the center gives

$$[\Omega^2 - 16GV^2]_{106} = 8A^8Y^8k^8U^6b^2 - 48A^6Y^{12}k^8U^6b^2. \tag{14}$$

The exact homogeneous component is meant here; these are not estimates from a numerical ray. The full checked coefficient array of the defect is included in the whole-product component.

The global polynomial identity

$$\begin{aligned} P_{208} = {} & (Q-1)Z_0^2 - 2(Q-1)M\Omega Z_0 \\ & + M^2\{Q(\Omega^2 - 16GV^2) - \Omega^2 + 16G\} \end{aligned} \tag{15}$$

has successive upper degrees $122, 177, 208, 152, 72$. In particular, using the square weight makes $\deg Z_0 = 21$; retaining bV here would make the cross term degree 213 and obscure the lower-degree norm defect.

Set

$$q_0 = \beta J, \qquad k_0 = \eta + \zeta, \qquad B_0 = k_0 sT - q_0^2(q_0 - F).$$

Substitution of the actual leading components in (14) gives

$$\boxed{[P_{208}]_{208} = 9\,2^{11} q_0^{88} w^{18} s^{52} i^{10} k_0^{32} B_0^2 (w^2 - 6s^2).} \tag{16}$$

This is nonzero over the polynomial ring for the actual $\beta > 0$. Thus the exact degree is 208. The factors B_0 and $w^2 - 6s^2$ must be retained; suppressing a tied leading term would give an incomplete degree certificate.

The complete literal circuit has 116 live binary operations: 71 multiplications, 24 additions and 21 subtractions. Every gate, input and fixed port is used. The independent checker binds its entire unchanged prefix to the frozen 85-operation parent, verifies the new center as a polynomial identity, checks the altered carrier and finalizer, and establishes the exact leading form. The reported count does not count a displayed abbreviation as a free arithmetic operation.

7 Direct quartics and nonnegativity on the zero set

The direct component is logically independent of the native coefficient tuple. Its source is package 164's C222 arithmetic construction, and its projection technique descends from packages 159 and 163 [3, 1]. The fresh content is the particular measured reductions and the natural-domain argument.

Profile	Domain	Variables	Rows	Terms	Height
Natural, principal	$\mathbb{N}$	203	155	8,237	799,349,232
Natural, lower height	$\mathbb{N}$	203	155	49,682	481,920
Natural, alternate basis	$\mathbb{N}$	203	155	8,263	207,116,893
Natural, intermediate	$\mathbb{N}$	204	156	8,213	130,686,374
Integer, principal	$\mathbb{Z}$	233	169	10,379	12,874
Integer, sparse	$\mathbb{Z}$	234	170	4,858	13,448

All six polynomials have exact degree four. Terms and height refer to the collected quartic; height is the largest absolute coefficient.

The 203-natural equation trades coefficient height against the smaller dimension; the row-basis variants make that tradeoff explicit. The 233-integer equation reduces the inspected 234-variable dimension record in this direct family, but does not improve every coefficient measure.

7.1 An exact quadratic projection

Let $q_1, \dots, q_m$ be integer quadratics. Suppose an integer vector v is in the common kernel of their homogeneous quadratic Hessians, with $v_p = 1$. In coordinates $x = y + vz$, $y_p = 0$, each row has the form

$$q_j(y + vz) = q_j(y) + a_j z, \qquad a_j \in \mathbb{Z}.$$

If $a_r = \pm 1$, eliminate $z = -a_r q_r(y)$. The remaining quadratics are

$$q_j(y) - a_r a_j q_r(y) = 0 \qquad (j \neq r).$$

These maps are inverse on unrestricted integer zero sets. The independent checker reconstructs the target by this row-combination identity; the constructor uses shear and substitution. Natural domains require an additional argument, which is not supplied by the Hessian calculation.

7.2 Two canonical sign choices remove two natural coordinates

In the pinned natural source, put

$$A_1 = \texttt{mul.363}, \quad B_1 = \texttt{mul.369}, \qquad Y_1 = B_1 - A_1.$$

The row $A_1 = (Sp + 1)(1 + \texttt{mul.361})$ restores $A_1 \geq 0$ and $B_1 = Y_1 + A_1 \geq 0$ from every target natural zero. The source's canonical CRT construction already chooses $B_1 > A_1$, supplying the reverse existential direction. The second pair is

$$C_1 = \texttt{mul.372}, \quad D_1 = \texttt{mul.373}, \qquad Y_2 = C_1 - D_1,$$

with $D_1 = (S + 1)(1 + \texttt{mul.361})$ and $C_1 = Y_2 + D_1$. The same canonical representatives satisfy $C_1 > D_1$. These reductions give 204 nonnegative variables. They are not claimed to be bijections on every arbitrary natural source zero.

7.3 A retained equation certifies the third restoration

Write $u = \texttt{mul.299}$, $v = \texttt{mul.304}$ and $t = \texttt{add.302}$. Their defining rows are $u = \Pi$ and $v = t(t + 1)$, where

$$\begin{aligned} \Pi = {} & 4556 + 4320r + 1024r^2 + 512rs + 128rt_1 \\ & + 1080s + 64s^2 + 32st_1 + 270t_1 + 4t_1^2, \\ r = \texttt{mul.232}, \quad & s = \texttt{mul.237}, \quad t_1 = \texttt{mul.243}. \end{aligned}$$

The common quadratic direction is $(u, v) \mapsto (u + z, v - 4z)$. Retain $w_1 = v + 4u$, eliminate u, and restore

$$u = \Pi, \qquad v = w_1 - 4\Pi.$$

Although the latter polynomial has negative coefficients, its retained row forces $w_1 - 4\Pi = t(t+1) \geq 0$ on every target natural zero. Conversely $w_1 = v + 4u \geq 0$ on every old natural zero. Thus this third step is a pointwise natural zero-set bijection and gives 203 variables.

The useful principle is that a restoration only has to be nonnegative *on the target zero set*. Requiring its coefficients all to be nonnegative would miss this elimination. The package verifies the exact retained row and the entire restoration, not just its expected shape.

8 What was investigated and what is not claimed

We investigated splitting the general C222 seed bundle into a shallow guarded vacuity axiom and a fixed-label seed core. This initially looked like a new semantic simplification. A source audit found exactly that proposal already measured in package 156, with negative size results [4]. An updated trial still did not beat the current natural equation, so it is not promoted as a new construction. This prevents an apparent change of notation or an old negative experiment from being reported as a breakthrough.

Whole-expression polynomial centers already occur in the eleven-unrestricted-integer branch of packages 153, 154 and 165 [6]. The new contribution here is the ten-positive-variable specialization: the explicit quotient center for EV, its majorant bound before recovery of the encoded signs, and the resulting literal degree-208 equation. Its unbounded proof is separate from the finite checks. This endpoint and its specific argument appear new relative to the inspected repository boundary; no exhaustive literature-priority claim is made.

There is no proof here that degree 208 is best for ten positive unknowns, that 203 or 233 variables are best among direct quartics, or that any one metric captures "simplest." The small-variable forms retain large fixed coefficients. The direct quartics have smaller coefficients and many more variables. A future formalization must check the new Diophantine bridge, not merely reuse the existing machine endpoint.

9 Reproducibility and integration

The ZIP contains this PDF and all native LaTeX sources; the ten-variable graphs; four authenticated fixed numerals and their finite reconstruction; the direct quadratic residual lists and collected quartic expansions; ordinary proofs; independent coefficient/certificate checkers; source ledgers, licenses, and recorded verification receipts.

The top-level command

```
python3 -B verify_all.py
```

checks the new finite artifacts using only the Python standard library. An optimized run is supported with `python3 -B -O verify_all.py`. The report build command is `sh build_report.sh`; it requires an ordinary TeX Live installation. Mathematical definitions are the authenticated graph and coefficient arrays, not PDF metadata.

The final archive is an additive submission suitable for `docs/incoming`. No remote repository content was modified. The provenance boundary and internal AI-assisted reviews are disclosed; they are not represented as outside human peer review.

A The complete fixed coefficient recipe

The following finite recipe is retained from the authenticated indexed native source. It is a specification of four fixed integers, independent of all quantified variables. Its bounded sums and fixed decoder do not introduce variable exponents into the final polynomial. The executable reconstructor verifies the exact raw bytes of all four integers. In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\,\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\,\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$

$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$

$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$

$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$

$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$

$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$

$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B The principal literal arithmetic circuit

The following table prints the entire straight-line evaluator. A row $z = x \circ y$ introduces a polynomial abbreviation; only the ten listed witnesses are quantified. The table is generated from, and checked against, the canonical JSON graph. The last register, `N`, is P_{208}.

Row	Register	Right-hand side
1	bJ	beta × J
2	q	bJ + 1
3	q2	q × q
4	q4	q2 × q2
5	q6	q4 × q2
6	sixq6	6 × q6
7	Y	q × s
8	qmF	q − F
9	C	qmF − alpha
10	X	w × q
11	k	eta + zeta
12	kY	k × Y
13	c	kY + eta
14	XY	X × Y
15	Am1	XY + Y
16	4Am1	4 × Am1
17	Hp	4Am1 − 1
18	Ap1	Am1 + 2
19	V0k	XY × kY
20	V0kk	V0k + k
21	V0k2	V0k × V0kk
22	Dtau	V0k2 + 1
23	G1	bJ × qmF
24	G	G1 + alpha
25	q2m1	q2 − 1
26	Gq	G × q2m1
27	qMF	q × MF
28	MCqMF	MC + qMF
29	RJ	MCqMF × J
30	R	Gq + RJ
31	Rp1	R + 1
32	e	k − Rp1
33	ksq	k × k
34	ksqm4	ksq − 4
35	inverse_cubic	V0k × ksqm4
36	q4s	q4 × s
37	e_sigma	e − q4s
38	six_e_sigma	6 × e_sigma
39	eclear	six_e_sigma − inverse_cubic
40	K0w	K0 + w
41	K0wC	K0w × C
42	Bt	K0wC − F
43	c2	c × c
44	U	i × c2
45	UE	U × Dtau
46	Delta	Am1 × Ap1
47	SE	Delta × UE
48	DpI2	SE × U
49	Df	DpI2 + 1
50	aF0	DpI2 × Df
51	fouraF0	4 × aF0
52	trace_f	DpI2 + Df
53	Q	Delta × fouraF0
54	cT	c × T
55	cTR	cT − R
56	DfcTR	trace_f × cTR
57	V	DfcTR − c
58	V2	V × V
59	P1mu2	bJ × sixq6
60	wp_A	Am1 + 1

Continued on next page

Row	Register	Right-hand side (continued)
61	wp_a2	kY × kY
62	wp_Yp1	Y + 1
63	wp_kYYp1	kY × wp_Yp1
64	wp_sinner	V0kk − wp_kYYp1
65	wp_S	kY × wp_sinner
66	wp_U2	U × U
67	wp_theta	wp_U2 × cTR
68	wp_threeE	3 × Dtau
69	wp_Zc	wp_threeE + wp_a2
70	wp_thetaZc	wp_theta × wp_Zc
71	wp_bmc	cTR − c
72	wp_Rc	wp_thetaZc − wp_bmc
73	wp_E2	Dtau × Dtau
74	wp_fourE2	4 × wp_E2
75	wp_S2	wp_S × wp_S
76	wp_threeS2	3 × wp_S2
77	wp_a22	wp_a2 × wp_a2
78	wp_sum1	wp_fourE2 + wp_threeS2
79	wp_Sc	wp_sum1 + wp_a22
80	wp_EV	Dtau × V
81	wp_fourEV	4 × wp_EV
82	wp_thetaSc	wp_theta × wp_Sc
83	wp_Vdiff	wp_fourEV − wp_thetaSc
84	wp_Apart	wp_A × wp_Vdiff
85	wp_SRc	wp_S × wp_Rc
86	wp_twoSRc	2 × wp_SRc
87	wp_inner	wp_Apart + wp_twoSRc
88	wp_W4	U × wp_inner
89	L	cTR × cTR
90	LBt	L × Bt
91	wp_raw_center	n − LBt
92	center	wp_raw_center − wp_W4
93	Hn	Hp × center
94	threec	3 × c
95	twoX	2 × X
96	gap	threec − twoX
97	Kpos0	X × gap
98	Kpos	Kpos0 + 2
99	lin	Hn + Kpos
100	Zpart1	P1mu2 × lin
101	mu2inner	sixq6 × Bt
102	P1e	bJ × eclear
103	brk	mu2inner + P1e
104	Zpart2	Hp × brk
105	Znum	Zpart1 − Zpart2
106	Mtot	P1mu2 × Hp
107	M2	Mtot × Mtot
108	Dg	Dtau × Df
109	wp_G16	16 × Dg
110	Z2	Znum × Znum
111	RG	M2 × wp_G16
112	RGV2	RG × V2
113	inside	Z2 − RGV2
114	Qinside	Q × inside
115	correction	RG − Z2
116	N	Qinside + correction

C Literal residuals of the 203-natural quartic

Every coordinate below is independently in $\mathbb{N}$. The equation is exactly $\sum_{j=0}^{154} q_j(x_0,\dots,x_{202})^2 = 0$. The complete expanded quartic in the archive is independently regenerated from these residuals. An index denotes the canonical coordinate order in `direct/natural203/equation.json`.

$$q_0 = 1 - x_3^2 + x_4$$

$$q_1 = 1 + x_2 - x_3 - x_3x_5 - x_3x_6 - x_5$$

$$q_2 = -2\,x_0x_2 - 5\,x_2 + x_9$$

$$q_3 = -1 + x_1 - x_2x_7 - x_3 - x_7$$

$$q_4 = -1 + x_1 - x_8 - x_8x_9$$

$$q_5 = 1 - x_4 - x_5x_{20} - x_6x_{20} + x_{18} - x_{20}$$

$$q_6 = -5\,x_{10} - x_{10}^2 - 2\,x_{10}x_{11} + 2\,x_{11}x_{12} + 5\,x_{12} + x_{12}^2 + x_{21} - x_{25}$$

$$q_7 = -16 - 8\,x_{11} - 4\,x_{21} + x_{22} - x_{35}$$

$$q_8 = -30 - 11\,x_{10} - x_{10}^2 - 4\,x_{10}x_{22} - 22\,x_{22} - 4\,x_{22}^2 + x_{23}$$

$$q_9 = -380 - 312\,x_{22} - 64\,x_{22}^2 - 32\,x_{22}x_{23} - 78\,x_{23} - 4\,x_{23}^2 + x_{24}$$

$$q_{10} = -6 - 5\,x_{11} - x_{11}^2 - 2\,x_{11}x_{12} - 5\,x_{12} - x_{12}^2 + x_{25}$$

$$q_{11} = -6 + 2\,x_{10}x_{11} - 2\,x_{10}x_{12} - 2\,x_{11}x_{12} - 10\,x_{12} - 2\,x_{12}^2 - x_{21} + x_{25} + x_{26}$$

$$q_{12} = -138 + 8\,x_{10} - 170\,x_{11} - 30\,x_{11}^2 - 48\,x_{11}x_{12} - 32\,x_{11}x_{21} - 24\,x_{11}x_{26} - 48\,x_{12} - 32\,x_{12}x_{21} - 100\,x_{21} - 8\,x_{21}^2 - 16\,x_{21}x_{26} + 16\,x_{25} - 40\,x_{26} + x_{27} - x_{31}$$

$$q_{13} = -240 - 32\,x_{10} - 128\,x_{12} - 64\,x_{25} - 6\,x_{27} - 2\,x_{27}^2 + x_{28} + 6\,x_{31} + 2\,x_{31}^2 - x_{32}$$

$$q_{14} = -14 - 4\,x_{10} - 16\,x_{10}x_{25} - 16\,x_{10}x_{26} - 4\,x_{11} - 16\,x_{11}x_{25} - 16\,x_{11}x_{26} - 16\,x_{21} - 42\,x_{25} - 4\,x_{25}^2 - 8\,x_{25}x_{26} - 42\,x_{26} - 4\,x_{26}^2 + x_{29}$$

$$q_{15} = -1592 - 2264\,x_{11} - 800\,x_{11}^2 - 160\,x_{11}x_{21} - 640\,x_{11}x_{25} - 160\,x_{11}x_{29} - 220\,x_{21} - 8\,x_{21}^2 - 64\,x_{21}x_{25} - 16\,x_{21}x_{29} - 912\,x_{25} - 128\,x_{25}^2 - 64\,x_{25}x_{29} - 228\,x_{29} - 8\,x_{29}^2 + x_{30}$$

$$
\begin{aligned}
q_{16} = {} & -180 + 10\,x_{10} + 6\,x_{10}^2 - 8\,x_{10}x_{25} \\
& + 10\,x_{11} + 6\,x_{11}^2 - 8\,x_{11}x_{26} - 192\,x_{12} \\
& - 48\,x_{12}^2 - 32\,x_{12}x_{25} - 32\,x_{12}x_{26} - 76\,x_{25} \\
& - 8\,x_{25}^2 - 76\,x_{26} - 8\,x_{26}^2 + x_{31}
\end{aligned}
$$

$$
\begin{aligned}
q_{17} = {} & 360 - 40\,x_{10} - 24\,x_{10}^2 + 32\,x_{10}x_{25} \\
& + 384\,x_{12} + 96\,x_{12}^2 + 128\,x_{12}x_{25} + 272\,x_{25} \\
& + 32\,x_{25}^2 + 32\,x_{26} - 6\,x_{31} - 2\,x_{31}^2 \\
& + x_{32}
\end{aligned}
$$

$$
\begin{aligned}
q_{18} = {} & -110 - 168\,x_{11} - 64\,x_{11}^2 - 32\,x_{11}x_{21} \\
& - 32\,x_{11}x_{25} - 42\,x_{21} - 4\,x_{21}^2 - 8\,x_{21}x_{25} \\
& - 42\,x_{25} - 4\,x_{25}^2 + x_{33}
\end{aligned}
$$

$$
\begin{aligned}
q_{19} = {} & -132 - 220\,x_{10} - 16\,x_{10}^2 - 128\,x_{10}x_{11} \\
& - 64\,x_{10}x_{21} - 16\,x_{10}x_{26} - 16\,x_{10}x_{33} - 80\,x_{11} \\
& - 64\,x_{11}x_{26} - 64\,x_{11}x_{33} - 40\,x_{21} - 32\,x_{21}x_{26} \\
& - 32\,x_{21}x_{33} - 110\,x_{26} - 4\,x_{26}^2 - 8\,x_{26}x_{33} \\
& - 110\,x_{33} - 4\,x_{33}^2 + x_{34} - 4\,x_{39}
\end{aligned}
$$

$$
\begin{aligned}
q_{20} = {} & -14 - 8\,x_{11} + 8\,x_{11}x_{12} + 13\,x_{12} \\
& + x_{12}^2 + 4\,x_{12}x_{21} - 4\,x_{21} + x_{35} \\
& - x_{61}
\end{aligned}
$$

$$
\begin{aligned}
q_{21} = {} & -30 - 11\,x_{10} - x_{10}^2 - 4\,x_{10}x_{35} \\
& - 22\,x_{35} - 4\,x_{35}^2 + x_{36}
\end{aligned}
$$

$$
\begin{aligned}
q_{22} = {} & -380 - 312\,x_{35} - 64\,x_{35}^2 - 32\,x_{35}x_{36} \\
& - 78\,x_{36} - 4\,x_{36}^2 + x_{37}
\end{aligned}
$$

$$
\begin{aligned}
q_{23} = {} & -30 - 11\,x_{11} - x_{11}^2 - 4\,x_{11}x_{37} \\
& - 22\,x_{37} - 4\,x_{37}^2 + x_{38}
\end{aligned}
$$

$$
\begin{aligned}
q_{24} = {} & -18 - 8\,x_{10} - 30\,x_{11} - 34\,x_{11}^2 \\
& + 48\,x_{11}x_{12} - 32\,x_{11}x_{21} + 24\,x_{11}x_{26} + 48\,x_{12} \\
& + 32\,x_{12}x_{21} - 8\,x_{21}^2 + 16\,x_{21}x_{26} - 16\,x_{25} \\
& + 40\,x_{26} - x_{27} + x_{31} + x_{39}
\end{aligned}
$$

$$
\begin{aligned}
q_{25} = {} & -782 - 700\,x_{11} - 144\,x_{11}^2 - 64\,x_{11}x_{21} \\
& - 80\,x_{11}x_{25} - 80\,x_{11}x_{39} - 200\,x_{21} - 32\,x_{21}x_{25} \\
& - 32\,x_{21}x_{39} - 150\,x_{25} - 4\,x_{25}^2 - 8\,x_{25}x_{39} \\
& - 154\,x_{39} - 4\,x_{39}^2 + x_{40}
\end{aligned}
$$

$$
\begin{aligned}
q_{26} = {} & -6272 - 1020\,x_{10} - 16\,x_{10}^2 - 512\,x_{10}x_{11} \\
& - 256\,x_{10}x_{21} - 16\,x_{10}x_{26} - 64\,x_{10}x_{39} - 16\,x_{10}x_{40} \\
& - 3520\,x_{11} - 256\,x_{11}x_{26} - 1024\,x_{11}x_{39} - 256\,x_{11}x_{40} \\
& - 1760\,x_{21} - 128\,x_{21}x_{26} - 512\,x_{21}x_{39} - 128\,x_{21}x_{40} \\
& - 510\,x_{26} - 4\,x_{26}^2 - 32\,x_{26}x_{39} - 8\,x_{26}x_{40} \\
& - 2104\,x_{39} - 64\,x_{39}^2 - 32\,x_{39}x_{40} - 510\,x_{40} \\
& - 4\,x_{40}^2 + x_{41}
\end{aligned}
$$

$$
\begin{aligned}
q_{27} = {} & -6 - 5\,x_{12} - x_{12}^2 - 2\,x_{12}x_{13} \\
& - 5\,x_{13} - x_{13}^2 + x_{42}
\end{aligned}
$$

$$
\begin{aligned}
q_{28} = & -224 - 26\,x_{10} - 2\,x_{10}^2 - 16\,x_{10}x_{13} \\
& - 8\,x_{10}x_{42} - 288\,x_{12} - 128\,x_{12}^2 - 64\,x_{12}x_{25} \\
& - 64\,x_{12}x_{26} - 88\,x_{13} - 32\,x_{13}^2 - 32\,x_{13}x_{42} \\
& - 76\,x_{25} - 8\,x_{25}^2 - 16\,x_{25}x_{26} - 68\,x_{26} \\
& - 8\,x_{26}^2 - 44\,x_{42} - 8\,x_{42}^2 + x_{43}
\end{aligned}
$$

$$
\begin{aligned}
q_{29} = & \,256 + 104\,x_{10} + 8\,x_{10}^2 + 64\,x_{10}x_{13} \\
& + 32\,x_{10}x_{42} + 352\,x_{13} + 128\,x_{13}^2 + 128\,x_{13}x_{42} \\
& + 176\,x_{42} + 32\,x_{42}^2 - 6\,x_{43} - 2\,x_{43}^2 \\
& + x_{44}
\end{aligned}
$$

$$
\begin{aligned}
q_{30} = & -42 - 52\,x_{12} - 16\,x_{12}^2 - 16\,x_{12}x_{25} \\
& - 26\,x_{25} - 4\,x_{25}^2 + x_{45}
\end{aligned}
$$

$$
\begin{aligned}
q_{31} = & -30 - 11\,x_{12} - x_{12}^2 - 4\,x_{12}x_{45} \\
& - 22\,x_{45} - 4\,x_{45}^2 + x_{46}
\end{aligned}
$$

$$
\begin{aligned}
q_{32} = & -306 - 35\,x_{11} - x_{11}^2 - 16\,x_{11}x_{45} \\
& - 4\,x_{11}x_{46} - 280\,x_{45} - 64\,x_{45}^2 - 32\,x_{45}x_{46} \\
& - 70\,x_{46} - 4\,x_{46}^2 + x_{47}
\end{aligned}
$$

$$
\begin{aligned}
q_{33} = & -4556 - 4320\,x_{45} - 1024\,x_{45}^2 - 512\,x_{45}x_{46} \\
& - 128\,x_{45}x_{47} - 1080\,x_{46} - 64\,x_{46}^2 - 32\,x_{46}x_{47} \\
& - 270\,x_{47} - 4\,x_{47}^2 + x_{48}
\end{aligned}
$$

$$
\begin{aligned}
q_{34} = & -30 - 11\,x_{12} - x_{12}^2 - 4\,x_{12}x_{48} \\
& - 22\,x_{48} - 4\,x_{48}^2 + x_{49}
\end{aligned}
$$

$$
\begin{aligned}
q_{35} = & -306 - 35\,x_{11} - x_{11}^2 - 16\,x_{11}x_{48} \\
& - 4\,x_{11}x_{49} - 280\,x_{48} - 64\,x_{48}^2 - 32\,x_{48}x_{49} \\
& - 70\,x_{49} - 4\,x_{49}^2 + x_{50}
\end{aligned}
$$

$$
\begin{aligned}
q_{36} = & -880 - 1280\,x_{11} - 480\,x_{11}^2 - 512\,x_{11}x_{25} \\
& - 128\,x_{11}x_{29} + 74\,x_{12} + 30\,x_{12}^2 - 32\,x_{12}x_{25} \\
& - 8\,x_{12}x_{29} - 752\,x_{25} - 128\,x_{25}^2 - 64\,x_{25}x_{29} \\
& - 180\,x_{29} - 8\,x_{29}^2 - 32\,x_{48} - 8\,x_{49} \\
& - 2\,x_{50} + x_{51}
\end{aligned}
$$

$$
\begin{aligned}
q_{37} = & -264 - 128\,x_{48} - 32\,x_{49} - 8\,x_{50} \\
& - 2\,x_{51} - 2\,x_{51}^2 + x_{52}
\end{aligned}
$$

$$
\begin{aligned}
q_{38} = & -2 - 3\,x_{10} - x_{10}^2 - 2\,x_{10}x_{52} \\
& - 3\,x_{52} - x_{52}^2 + x_{53}
\end{aligned}
$$

$$
\begin{aligned}
q_{39} = & -12 - 28\,x_{52} - 16\,x_{52}^2 - 16\,x_{52}x_{53} \\
& - 14\,x_{53} - 4\,x_{53}^2 + x_{54}
\end{aligned}
$$

$$
\begin{aligned}
q_{40} = & -30 - 11\,x_{12} - x_{12}^2 - 4\,x_{12}x_{54} \\
& - 22\,x_{54} - 4\,x_{54}^2 + x_{55}
\end{aligned}
$$

$$
\begin{aligned}
q_{41} = & -306 - 35\,x_{10} - x_{10}^2 - 16\,x_{10}x_{54} \\
& - 4\,x_{10}x_{55} - 280\,x_{54} - 64\,x_{54}^2 - 32\,x_{54}x_{55} \\
& - 70\,x_{55} - 4\,x_{55}^2 + x_{56}
\end{aligned}
$$

$$
\begin{aligned}
q_{42} = & -4556 - 4320\,x_{54} - 1024\,x_{54}^2 - 512\,x_{54}x_{55} \\
& - 128\,x_{54}x_{56} - 1080\,x_{55} - 64\,x_{55}^2 - 32\,x_{55}x_{56} \\
& - 270\,x_{56} - 4\,x_{56}^2 + x_{57}
\end{aligned}
$$

$$\begin{aligned} q_{43} = & -48 - 24\,x_{12} - 16\,x_{12}x_{57} - 12\,x_{25} \\ & - 8\,x_{25}x_{57} - x_{45} - 38\,x_{57} - 4\,x_{57}^2 \\ & + x_{58} \end{aligned}$$

$$\begin{aligned} q_{44} = & -9794 - 70\,x_{11} - 2\,x_{11}^2 - 32\,x_{11}x_{57} \\ & - 8\,x_{11}x_{58} - 8640\,x_{48} - 2048\,x_{48}^2 - 1024\,x_{48}x_{49} \\ & - 256\,x_{48}x_{50} - 2160\,x_{49} - 128\,x_{49}^2 - 64\,x_{49}x_{50} \\ & - 540\,x_{50} - 8\,x_{50}^2 - 592\,x_{57} - 128\,x_{57}^2 \\ & - 64\,x_{57}x_{58} - 148\,x_{58} - 8\,x_{58}^2 + x_{59} \end{aligned}$$

$$\begin{aligned} q_{45} = & -18224 - 17280\,x_{48} - 4096\,x_{48}^2 - 2048\,x_{48}x_{49} \\ & - 512\,x_{48}x_{50} - 4320\,x_{49} - 256\,x_{49}^2 - 128\,x_{49}x_{50} \\ & - 1080\,x_{50} - 16\,x_{50}^2 - x_{59} - x_{59}^2 \\ & + x_{60} \end{aligned}$$

$$\begin{aligned} q_{46} = & -30 - 42\,x_{11} - 14\,x_{11}^2 - 4\,x_{11}x_{12} \\ & - 16\,x_{11}x_{21} - 3\,x_{12} + x_{12}^2 - 4\,x_{12}x_{21} \\ & - 26\,x_{21} - 4\,x_{21}^2 - 2\,x_{25} + x_{61} \end{aligned}$$

$$\begin{aligned} q_{47} = & \,522 + 736\,x_{11} - 160\,x_{11}x_{61} - 70\,x_{12} \\ & - 2\,x_{12}^2 - 8\,x_{12}x_{60} + 368\,x_{21} - 80\,x_{21}x_{61} \\ & - 50\,x_{35} - 148\,x_{60} - 8\,x_{60}^2 - 228\,x_{61} \\ & - 8\,x_{61}^2 + x_{62} \end{aligned}$$

$$\begin{aligned} q_{48} = & -4800 - 2944\,x_{11} + 640\,x_{11}x_{61} - 1472\,x_{21} \\ & + 320\,x_{21}x_{61} + 200\,x_{35} + 912\,x_{61} + 32\,x_{61}^2 \\ & - 6\,x_{62} - 2\,x_{62}^2 + x_{63} \end{aligned}$$

$$q_{49} = -2 - 3\,x_{63} - x_{63}^2 + x_{64}$$

$$q_{50} = -x_3 - x_3x_{15} + x_{65}$$

$$q_{51} = -2\,x_{24}x_{65} - 4\,x_{65} + x_{66}$$

$$q_{52} = -x_{28}x_{65} + x_{67}$$

$$q_{53} = -x_{30}x_{65} + x_{68}$$

$$q_{54} = -x_{32}x_{65} + x_{69}$$

$$\begin{aligned} q_{55} = & -64\,x_{11}x_{65} - 32\,x_{21}x_{65} - 8\,x_{33}x_{65} - 2\,x_{34}x_{65} \\ & - 80\,x_{65} + x_{70} \end{aligned}$$

$$q_{56} = -4\,x_{11}x_{65} - 2\,x_{38}x_{65} - 4\,x_{65} + x_{71}$$

$$\begin{aligned} q_{57} = & -256\,x_{11}x_{65} - 128\,x_{21}x_{65} - 32\,x_{39}x_{65} - 8\,x_{40}x_{65} \\ & - 2\,x_{41}x_{65} - 480\,x_{65} + x_{72} \end{aligned}$$

$$q_{58} = -x_{44}x_{65} + x_{73}$$

$$q_{59} = -2\,x_{64}x_{65} - 4\,x_{65} + x_{74}$$

$$q_{60} = -x_3 - x_3x_{11} + x_{75}$$

$$q_{61} = -x_3 - x_3x_{10} + x_{76}$$

$$q_{62} = 2\,x_3 + x_{77} - x_{80}$$

$$\begin{aligned} q_{63} = & -1 - x_{14} - x_{14}x_{76} - x_{76} \\ & + x_{78} \end{aligned}$$

$$\begin{aligned} q_{64} = & \,1 + x_{15} + x_{15}x_{75} + x_{75} \\ & - x_{77}x_{78} - x_{78} + x_{79} \end{aligned}$$

$$q_{65} = -6\,x_3 - 4\,x_3x_{11} - 2\,x_3x_{21} + x_{80}$$

$$\begin{aligned} q_{66} = {} & x_{14} + x_{14}x_{75} - x_{15} - x_{15}x_{80} \\ & + x_{75} - x_{80} + x_{81} \end{aligned}$$

$$q_{67} = -2\,x_2 - x_2x_{18} + x_{82}$$

$$\begin{aligned} q_{68} = {} & -1 - x_{16} - x_{16}x_{82} - x_{82} \\ & + x_{83} \end{aligned}$$

$$\begin{aligned} q_{69} = {} & -1 - x_2 - x_2x_{17} - x_{17} \\ & - x_{17}x_{82} - x_{82} + x_{84} \end{aligned}$$

$$\begin{aligned} q_{70} = {} & 2 - 4\,x_1 + 2\,x_1^2 - 2\,x_1x_{14} \\ & - 2\,x_1x_{15} - 2\,x_1x_{83} - 2\,x_1x_{84} + 2\,x_{14} \\ & + x_{14}^2 + 2\,x_{14}x_{83} + 2\,x_{15} + x_{15}^2 \\ & + 2\,x_{15}x_{84} - x_{19}x_{85} + 2\,x_{83} + x_{83}^2 \\ & + 2\,x_{84} + x_{84}^2 + x_{201}^2 + x_{202}^2 \end{aligned}$$

$$\begin{aligned} q_{71} = {} & -x_{10}x_{19} - x_{11}x_{19} - x_{12}x_{19} - x_{13}x_{19} \\ & - x_{14}x_{19} - x_{15}x_{19} - x_{16}x_{19} - x_{17}x_{19} \\ & - x_{18}x_{19} - 10\,x_{19} + x_{86} \end{aligned}$$

$$\begin{aligned} q_{72} = {} & 1 - x_{87} - x_{87}x_{132} - x_{87}x_{133} \\ & - x_{87}x_{183} + x_{138} + x_{139} + x_{145} \\ & + x_{150} + x_{155} + x_{186} \end{aligned}$$

$$\begin{aligned} q_{73} = {} & -x_{19} - x_{19}x_{172} - x_{19}x_{173} - x_{19}x_{178} \\ & + x_{88} \end{aligned}$$

$$\begin{aligned} q_{74} = {} & -34 - 2\,x_0 - 2\,x_0x_{91} - x_1 \\ & - x_1x_{91} - 2\,x_2 - 2\,x_2x_{91} - x_4 \\ & - x_4x_{91} - x_5 - x_5x_{91} - x_6 \\ & - x_6x_{91} - x_{10} - 2\,x_{10}x_{91} - x_{11} \\ & - 334\,x_{11}x_{91} - x_{12} - x_{12}x_{91} - x_{13} \\ & - x_{13}x_{91} - x_{14} - x_{14}x_{91} - x_{15} \\ & - x_{15}x_{91} - x_{16} - x_{16}x_{91} - x_{17} \\ & - x_{17}x_{91} - x_{18} - x_{18}x_{91} - x_{19} \\ & - x_{19}x_{91} - 164\,x_{21}x_{91} - 2\,x_{24}x_{91} - x_{28}x_{91} \\ & - x_{30}x_{91} - x_{32}x_{91} - 8\,x_{33}x_{91} - 2\,x_{34}x_{91} \\ & - 2\,x_{38}x_{91} - 32\,x_{39}x_{91} - 8\,x_{40}x_{91} - 2\,x_{41}x_{91} \\ & - x_{44}x_{91} - 2\,x_{64}x_{91} - x_{86} - x_{86}x_{91} \\ & - x_{87} - x_{87}x_{91} - 619\,x_{91} - x_{91}x_{132} \\ & - x_{91}x_{133} - x_{91}x_{160} - x_{91}x_{161} - x_{91}x_{167} \\ & - x_{91}x_{172} - x_{91}x_{173} - x_{91}x_{178} - x_{91}x_{183} \\ & + x_{92} + 4\,x_{96}x_{107} - x_{98} - x_{132} \\ & - x_{133} - x_{160} - x_{161} - x_{167} \\ & - x_{172} - x_{173} - x_{178} - x_{183} \end{aligned}$$

$$
\begin{aligned}
q_{75} = \; & -2\,x_0x_{92} - x_1x_{92} - 2\,x_2x_{92} - x_4x_{92} \\
& - x_5x_{92} - x_6x_{92} - 2\,x_{10}x_{92} - 334\,x_{11}x_{92} \\
& - x_{12}x_{92} - x_{13}x_{92} - x_{14}x_{92} - x_{15}x_{92} \\
& - x_{16}x_{92} - x_{17}x_{92} - x_{18}x_{92} - x_{19}x_{92} \\
& - 164\,x_{21}x_{92} - 2\,x_{24}x_{92} - x_{28}x_{92} - x_{30}x_{92} \\
& - x_{32}x_{92} - 8\,x_{33}x_{92} - 2\,x_{34}x_{92} - 2\,x_{38}x_{92} \\
& - 32\,x_{39}x_{92} - 8\,x_{40}x_{92} - 2\,x_{41}x_{92} - x_{44}x_{92} \\
& - 2\,x_{64}x_{92} - x_{86}x_{92} - x_{87}x_{92} - 620\,x_{92} \\
& - x_{92}x_{132} - x_{92}x_{133} - x_{92}x_{160} - x_{92}x_{161} \\
& - x_{92}x_{167} - x_{92}x_{172} - x_{92}x_{173} - x_{92}x_{178} \\
& - x_{92}x_{183} + x_{93}
\end{aligned}
$$

$$
q_{76} = -1 + x_{89}^2 + x_{92}^2 - x_{93}^2
$$

$$
\begin{aligned}
q_{77} = \; & -1 - 2\,x_{87} - x_{88}x_{94} + 2\,x_{89} \\
& + 2\,x_{90} + x_{121}
\end{aligned}
$$

$$
q_{78} = -x_{97}^2 + x_{188}
$$

$$
\begin{aligned}
q_{79} = \; & 1 - x_{89} + x_{89}x_{188} - x_{90} \\
& + x_{90}x_{188} - 4\,x_{96}x_{188} + x_{99} - x_{188}
\end{aligned}
$$

$$
q_{80} = -x_{89}x_{96} - x_{90}x_{96} + x_{100}
$$

$$
q_{81} = -1 + x_{95}^2 + x_{96}^2 - x_{100}^2
$$

$$
q_{82} = -2\,x_{89}x_{105} - 2\,x_{90}x_{105} + x_{101}
$$

$$
q_{83} = -1 + x_{97}^2 - x_{101}^2 + 4\,x_{105}^2
$$

$$
q_{84} = -x_{98} - x_{98}x_{99} + x_{102}
$$

$$
\begin{aligned}
q_{85} = \; & -1 + x_{95}^2 + 2\,x_{95}x_{108} + x_{98}^2 \\
& - x_{102}^2 + x_{108}^2
\end{aligned}
$$

$$
q_{86} = -x_{96}^2 + x_{104}
$$

$$
q_{87} = -x_{103}x_{104} - x_{104} + x_{105}
$$

$$
q_{88} = -x_{97}x_{106} + x_{108}
$$

$$
\begin{aligned}
q_{89} = \; & -585 - x_{10} - 333\,x_{11} - 164\,x_{21} \\
& - 2\,x_{24} - x_{28} - x_{30} - x_{32} \\
& - 8\,x_{33} - 2\,x_{34} - 2\,x_{38} - 32\,x_{39} \\
& - 8\,x_{40} - 2\,x_{41} - x_{44} - 2\,x_{64} \\
& - 4\,x_{96}x_{107} + x_{98}
\end{aligned}
$$

$$
q_{90} = -x_{109}x_{110} + x_{111}
$$

$$
\begin{aligned}
q_{91} = \; & -1 - 2\,x_{89}x_{111} - 2\,x_{90}x_{111} + x_{109}^2 \\
& + x_{110}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{92} = \; & -1 - x_4^2 + 2\,x_4x_{89} + 2\,x_4x_{90} \\
& - x_{112}
\end{aligned}
$$

$$
\begin{aligned}
q_{93} = \; & 1 + x_2 - x_4x_{109} + x_{110} \\
& + x_{112}x_{113}
\end{aligned}
$$

$$
q_{94} = -x_{114}x_{115} + x_{116}
$$

$$
\begin{aligned}
q_{95} = \; & -1 - 2\,x_{89}x_{116} - 2\,x_{90}x_{116} + x_{114}^2 \\
& + x_{115}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{96} = {} & -2 - 2\,x_5 - x_5^2 - 2\,x_5x_6 \\
& + 2\,x_5x_{89} + 2\,x_5x_{90} - 2\,x_6 - x_6^2 \\
& + 2\,x_6x_{89} + 2\,x_6x_{90} + 2\,x_{89} + 2\,x_{90} \\
& - x_{117}
\end{aligned}
$$

$$
\begin{aligned}
q_{97} = {} & -x_5x_{114} - x_6x_{114} + x_{19} - x_{114} \\
& + x_{115} + x_{117}x_{118}
\end{aligned}
$$

$$
\begin{aligned}
q_{98} = {} & -1 - x_{86}^2 + 2\,x_{86}x_{89} + 2\,x_{86}x_{90} \\
& - x_{119}
\end{aligned}
$$

$$
q_{99} = x_{86}x_{114} - x_{87} - x_{115} - x_{119}x_{120}
$$

$$
\begin{aligned}
q_{100} = {} & 1 + x_{87}^2 - 2\,x_{87}x_{89} - 2\,x_{87}x_{90} \\
& + x_{121}
\end{aligned}
$$

$$
\begin{aligned}
q_{101} = {} & -1 + x_{87}x_{109} - x_{110} - x_{121}x_{122} \\
& - x_{132} - x_{133} - x_{183}
\end{aligned}
$$

$$
q_{102} = -x_{123}x_{124} + x_{125}
$$

$$
\begin{aligned}
q_{103} = {} & -1 - 2\,x_{89}x_{125} - 2\,x_{90}x_{125} + x_{123}^2 \\
& + x_{124}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{104} = {} & -1 + x_{87}x_{123} - x_{121}x_{126} - x_{124} \\
& - x_{160} - x_{161} - x_{167}
\end{aligned}
$$

$$
\begin{aligned}
q_{105} = {} & 1 - x_{87}x_{114} + x_{115} + x_{121}x_{127} \\
& + x_{172} + x_{173} + x_{178}
\end{aligned}
$$

$$
q_{106} = -x_{128}x_{129} + x_{130}
$$

$$
\begin{aligned}
q_{107} = {} & -1 - 2\,x_{89}x_{130} - 2\,x_{90}x_{130} + x_{128}^2 \\
& + x_{129}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{108} = {} & 1 + x_5 + x_5x_{132} + x_5x_{133} \\
& + x_5x_{183} + x_6 + x_6x_{132} + x_6x_{133} \\
& + x_6x_{183} - x_{87}x_{128} - x_{128} + x_{129} \\
& + x_{131} + x_{131}x_{138} + x_{131}x_{139} + x_{131}x_{145} \\
& + x_{131}x_{150} + x_{131}x_{155} + x_{131}x_{186} + 2\,x_{132} \\
& + x_{133} + x_{183}
\end{aligned}
$$

$$
q_{109} = -x_{134}x_{135} + x_{136}
$$

$$
\begin{aligned}
q_{110} = {} & -1 - 2\,x_{89}x_{136} - 2\,x_{90}x_{136} + x_{134}^2 \\
& + x_{135}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{111} = {} & -x_{19} - x_{19}x_{138} - x_{19}x_{139} - x_{19}x_{145} \\
& - x_{19}x_{150} - x_{19}x_{155} - x_{19}x_{186} + x_{140}
\end{aligned}
$$

$$
\begin{aligned}
q_{112} = {} & x_{87}x_{134} + x_{134} - x_{135} - x_{137}x_{140} \\
& - x_{138}
\end{aligned}
$$

$$
q_{113} = -x_{141}x_{142} + x_{143}
$$

$$
\begin{aligned}
q_{114} = {} & -1 - 2\,x_{89}x_{143} - 2\,x_{90}x_{143} + x_{141}^2 \\
& + x_{142}^2
\end{aligned}
$$

$$
\begin{aligned}
q_{115} = {} & -x_{87}x_{141} + x_{140}x_{144} - x_{141} + x_{142} \\
& + x_{145}
\end{aligned}
$$

$$
q_{116} = -x_{146}x_{147} + x_{148}
$$

$$q_{117} = -1 - 2\,x_{89}x_{148} - 2\,x_{90}x_{148} + x_{146}^2 + x_{147}^2$$

$$q_{118} = -x_{87}x_{146} + x_{140}x_{149} - x_{146} + x_{147} + x_{150}$$

$$q_{119} = -x_{151}x_{152} + x_{153}$$

$$q_{120} = -1 - 2\,x_{89}x_{153} - 2\,x_{90}x_{153} + x_{151}^2 + x_{152}^2$$

$$q_{121} = -x_{87}x_{151} + x_{140}x_{154} - x_{151} + x_{152} + x_{155}$$

$$q_{122} = -x_{156}x_{157} + x_{158}$$

$$q_{123} = -1 - 2\,x_{89}x_{158} - 2\,x_{90}x_{158} + x_{156}^2 + x_{157}^2$$

$$q_{124} = -x_{19} - x_{19}x_{160} - x_{19}x_{161} - x_{19}x_{167} + x_{162}$$

$$q_{125} = x_{87}x_{156} + x_{156} - x_{157} - x_{159}x_{162} - x_{160}$$

$$q_{126} = -x_{163}x_{164} + x_{165}$$

$$q_{127} = -1 - 2\,x_{89}x_{165} - 2\,x_{90}x_{165} + x_{163}^2 + x_{164}^2$$

$$q_{128} = -x_{87}x_{163} + x_{162}x_{166} - x_{163} + x_{164} + x_{167}$$

$$q_{129} = -x_{168}x_{169} + x_{170}$$

$$q_{130} = -1 - 2\,x_{89}x_{170} - 2\,x_{90}x_{170} + x_{168}^2 + x_{169}^2$$

$$q_{131} = -x_{87}x_{168} + x_{88}x_{171} - x_{168} + x_{169} + x_{172}$$

$$q_{132} = -x_{174}x_{175} + x_{176}$$

$$q_{133} = -1 - 2\,x_{89}x_{176} - 2\,x_{90}x_{176} + x_{174}^2 + x_{175}^2$$

$$q_{134} = x_{87}x_{174} - x_{88}x_{177} + x_{174} - x_{175} - x_{178}$$

$$q_{135} = -x_{179}x_{180} + x_{181}$$

$$q_{136} = -1 - 2\,x_{89}x_{181} - 2\,x_{90}x_{181} + x_{179}^2 + x_{180}^2$$

$$q_{137} = -x_{19} - x_{19}x_{132} - x_{19}x_{133} - x_{19}x_{183} + x_{184}$$

$$q_{138} = x_{87}x_{179} + x_{179} - x_{180} - x_{182}x_{184} - x_{183}$$

$$q_{139} = x_{87}x_{96} + x_{95} + x_{96} - x_{100} - x_{140}x_{185} - x_{186}$$

$$
\begin{aligned}
q_{140} = \; & -26 - 16\,x_{0} - 4\,x_{0}^{2} + 4\,x_{0}x_{109} \\
& - x_{1}^{2} + 2\,x_{1}x_{114} - 2\,x_{2} - x_{2}^{2} \\
& + 2\,x_{2}x_{123} - x_{4}^{2} + 2\,x_{4}x_{128} - 2\,x_{10} \\
& - x_{10}^{2} + 2\,x_{10}x_{134} - 2\,x_{11} - x_{11}^{2} \\
& + 2\,x_{11}x_{141} - 2\,x_{12} - x_{12}^{2} + 2\,x_{12}x_{146} \\
& - 2\,x_{13} - x_{13}^{2} + 2\,x_{13}x_{151} - 2\,x_{14} \\
& - x_{14}^{2} + 2\,x_{14}x_{156} - 2\,x_{15} - x_{15}^{2} \\
& + 2\,x_{15}x_{163} - 2\,x_{16} - x_{16}^{2} + 2\,x_{16}x_{168} \\
& - 2\,x_{17} - x_{17}^{2} + 2\,x_{17}x_{174} - 2\,x_{18} \\
& - x_{18}^{2} + 2\,x_{18}x_{179} - x_{89}x_{187} - x_{90}x_{187} \\
& + x_{96} + 4\,x_{96}x_{107} - x_{98} + 8\,x_{109} \\
& - x_{109}^{2} - x_{114}^{2} + 2\,x_{123} - x_{123}^{2} \\
& - x_{128}^{2} + 2\,x_{134} - x_{134}^{2} + 2\,x_{141} \\
& - x_{141}^{2} + 2\,x_{146} - x_{146}^{2} + 2\,x_{151} \\
& - x_{151}^{2} + 2\,x_{156} - x_{156}^{2} + 2\,x_{163} \\
& - x_{163}^{2} + 2\,x_{168} - x_{168}^{2} + 2\,x_{174} \\
& - x_{174}^{2} + 2\,x_{179} - x_{179}^{2} + x_{187}
\end{aligned}
$$

$$
\begin{aligned}
q_{141} = \; & - x_{14}^{2} + 2\,x_{14}x_{15} + x_{14}x_{66} - x_{15}^{2} \\
& - x_{15}x_{66} + x_{189}
\end{aligned}
$$

$$q_{142} = x_{14}x_{189} - x_{15}x_{189} - x_{67}x_{189} + x_{190}$$

$$q_{143} = x_{14}x_{190} - x_{15}x_{190} - x_{68}x_{190} + x_{191}$$

$$q_{144} = x_{14}x_{191} - x_{15}x_{191} - x_{69}x_{191} + x_{192}$$

$$q_{145} = x_{14}x_{192} - x_{15}x_{192} - x_{70}x_{192} + x_{193}$$

$$q_{146} = x_{14}x_{193} - x_{15}x_{193} - x_{71}x_{193} + x_{194}$$

$$q_{147} = x_{14}x_{194} - x_{15}x_{194} - x_{72}x_{194} + x_{195}$$

$$q_{148} = x_{14}x_{195} - x_{15}x_{195} - x_{73}x_{195} + x_{196}$$

$$q_{149} = - x_{79}x_{196} + x_{197}$$

$$q_{150} = - x_{81}x_{197} + x_{198}$$

$$
\begin{aligned}
q_{151} = \; & - x_{10}x_{11} + x_{10}x_{12} + x_{11}x_{12} - x_{12}^{2} \\
& + x_{199}
\end{aligned}
$$

$$q_{152} = x_{13}x_{198} + 2\,x_{198} - x_{198}x_{199} + x_{200}$$

$$q_{153} = - x_{13}x_{200} - x_{199}x_{200} - 2\,x_{200} + x_{201}$$

$$q_{154} = x_{14}x_{198} - x_{15}x_{198} - x_{74}x_{198} + x_{202}$$

References

[1] ProveIt, package 163, *Lower degree and fewer witnesses for Diophantine consistency*, 25 October 2026. In particular `compact/PROOF.md`, sections 3–9. https://github.com/VladimirReshetnikov/ProveIt/tree/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/163-trace-radical.

[2] ProveIt, package 164, *Unsquared factors and linear selectors for Diophantine consistency*, 25 October 2026. The positive inverse-Lucas proof and the natural canonical CRT proof are the immediate arithmetic parents. https://github.com/VladimirReshetnikov/ProveIt/tree/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/164-unsquared-factors-linear-selectors.

[3] ProveIt, package 159, *A direct quartic for Con(ZFC) in 234 integer variables*, 25 October 2026. Common-Hessian projection and independent algebraic certificates. https://github.com/VladimirReshetnikov/ProveIt/tree/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/159-quadratic-projection-234.

[4] ProveIt, package 156, *Integral projection lattice*, `research_notes/EXPERIMENT_REVIEW.md`. Prior measured seed/vacuity split. https://github.com/VladimirReshetnikov/ProveIt/blob/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/156-integral-projection-lattice/research_notes/EXPERIMENT_REVIEW.md.

[5] ProveIt, package 152, *Lucas corrections and elimination of arithmetic witnesses*, 25 October 2026. The earlier forward Lucas correction, retained in the source closure. https://github.com/VladimirReshetnikov/ProveIt/tree/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/152-lucas-corrections.

[6] ProveIt, packages 153, 154 and 165. Earlier whole-expression polynomial centers for eleven unrestricted integer variables: `153-nearest-factorial/signed/PROOF.md`, section 6; `154-normalized-residues-round19/report/scalar.tex`; and `165-wider-budgets-round20/scalar/PROOF.md`. https://github.com/VladimirReshetnikov/ProveIt/blob/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms/165-wider-budgets-round20/scalar/PROOF.md.

[7] ProveIt, `KEndpoint.lean`, `KSeedDirect222.lean`, `ConstructibleChoice.lean`. Named C222/ZF/ZFC interfaces, with the exact selected source copies and commit identifiers recorded in the artifact ledgers.

[8] J. P. Jones, *Universal Diophantine Equation*, Journal of Symbolic Logic 47 (1982), 549–571. https://doi.org/10.2307/2273588. Background for comparison only; the direct quartic component does not use this universal equation.